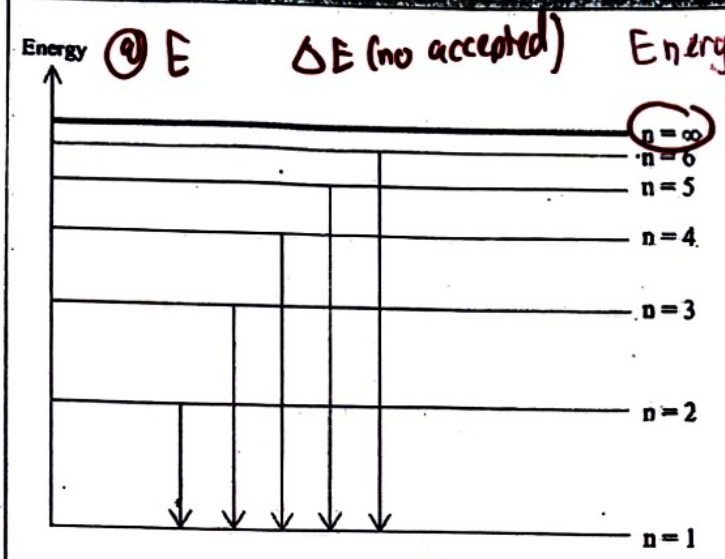
NO.	PART	SCHEME
1	(a)	% abundance $^{24}Mg = 78.99$ % abundance $^{25}Mg = x$ % abundance $^{26}Mg = y$ $78.99 + x + y = 100$ $y = 21.01 - x$ / must have Average atomic mass $= \frac{\Sigma Q_i M_i}{\Sigma Q_i}$ @ $24.3 = \frac{(78.99 \times 23.985) + (x)(24.986) + (21.01 - x)(25.983)}{100}$ $x = 10.51$ $y = 21.01 - x$ $= 21.01 - 10.51$ $= 10.50$ / m3 % abundance $^{25}Mg = 10.51$ @ 10.52 % abundance $^{26}Mg = 10.50$ @ 10.49
	(b)	$SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e$ / m1 $Br_2 + 2e \rightarrow 2Br^-$ / m7 $SO_2 + Br_2 + 2H_2O \rightarrow SO_4^{2-} + 2Br^- + 4H^+$ / m8
	(c)(i)	Assume volume of solution = 1 L Molarity $= \frac{n_{solute}}{V_{solution}}$ @ $0.2 \text{ mol L}^{-1} = \frac{n_{solute}}{1 \text{ L}}$ $n_{solute} = 0.2$ mol / m9 $m_{solute} = 0.2 \text{ mol} \times 40 \text{ g mol}^{-1} = 8.0$ g / m10 In 250 mL, $m_{solute} = 2.0$ g unit insist **Alternative:** $n_{solute} = MV = (0.2)\left(\frac{250}{1000}\right) = 0.05$ mol $m_{solute} = 0.05 \text{ mol} \times 40 \text{ g mol}^{-1}$ $= 2.0$ g unit insist — m10 ecf

m10 ecf if mol salah

m10 ecf

NO.	BAHAGIAN	SCHEME
	(c)(ii)	$M_1V_1 = M_2V_2$ @ $(0.2\ M)(250\ mL) = (0.1\ M)\ (V_2)$ $V_2 = 500\ mL$ M22 new addition Volume of $H_2O = (500 - 250)\ mL$ $= 250\ mL$ unit inst
	(d)(i)	$O_3 + NO \rightarrow O_2 + NO_2$ 0.015 accepted $n_{O_3} = \frac{0.740\ g}{48\ g\ mol^{-1}} = 0.0154\ mol$ 0.022 $n_{NO} = \frac{0.670\ g}{30\ g\ mol^{-1}} = 0.0223\ mol$ From equation; 1 mol of O_3 ≡ 1 mol of NO 0.0154 mol of O_3 ≡ **0.0154 mol of NO (needed)** m17 comparison **Since n_{NO} given > n_{NO} needed** m18 **hence NO is excess reactant while O_3 is limiting reactant.** m19 (correct reactant) **Alternative:** 0.015 $n_{O_3} = \frac{0.740\ g}{48\ g\ mol^{-1}} = 0.0154\ mol$ 0.022 $n_{NO} = \frac{0.670\ g}{30\ g\ mol^{-1}} = 0.0223\ mol$ 1 mol O_3 produces 1 mol NO_2 Thus, **0.0154 mol NO_2** is produced and 1 mol NO produces 1 mol NO_2 Thus, **0.0223 mol NO_2** is produced **Since O_3 produce less NO_2** **hence O_3 is the limiting reagent.**
	(d)(ii)	0.015 only mol lim NO_2 produced = 0.0154 mol × 46 g mol^{-1} m20 = **0.7084 g @ 0.709 g** m21 0.69 must 2 inst ecf Bsalah m20
		TOTAL

NO.	PART	SCHEME
2	(a)	



@ E ΔE (no accepted) Energy level X

kalau X & pax

Axis (energy and energy level)
Line spacing (energy level getting close as energy increases)
Arrow transition

all arrow correct

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad @$$

$$\frac{1}{\lambda} = 1.097 \times 10^7\left(\frac{1}{1^2} - \frac{1}{6^2}\right)$$

M5

if $n_1 < n_2$ any number accepted f

$$\lambda = 9.3762 \times 10^{-8}\ \text{m} \quad @ \quad 93.762\ \text{nm}$$

Alternative:

$$\Delta E = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

$$\Delta E = 2.18 \times 10^{-18}\left(\frac{1}{6^2} - \frac{1}{1^2}\right)$$

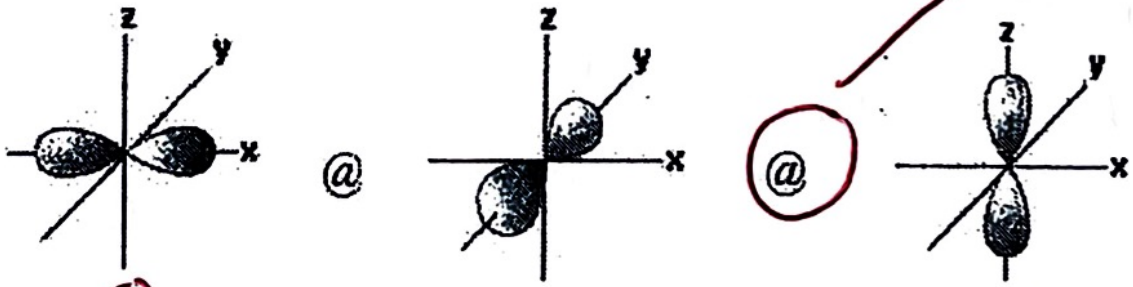
$$\Delta E = -2.12 \times 10^{-18}\ \text{J}$$

$$\Delta E = \frac{hc}{\lambda}$$

must have both eq.

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.6256 \times 10^{-34}\ \text{J s})(3.0 \times 10^8\ \text{m s}^{-1})}{2.12 \times 10^{-18}}$$

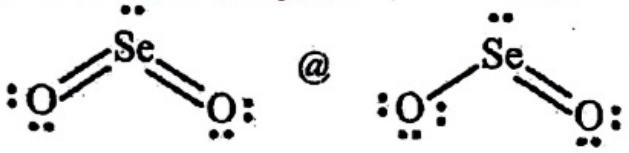
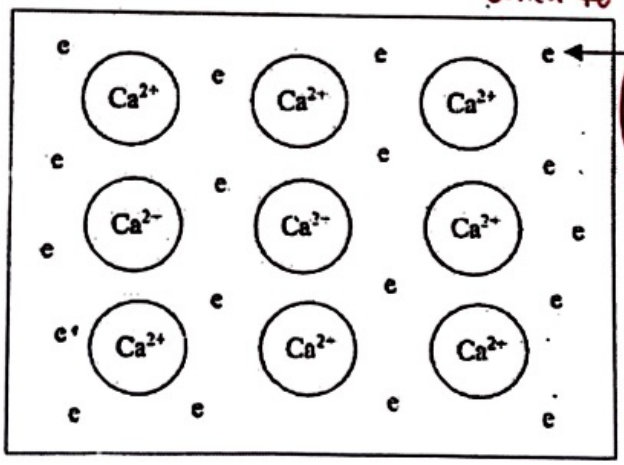
$$\lambda = 9.375 \times 10^{-8}\ \text{m}$$

JAWAPAN	MARKS	SKEMA
	(b)	$_{13}X$: $1s^2 2s^2 2p^6 3s^2 3p^1$ Orbital diagram (full green) also accepted When $\ell = 0 \rightarrow$ s orbital Thus, number of electrons in s orbital = **6** Orbital of valence electrons with $\ell = 1$ are 3p orbitals. kalau x c -1 lorek not incorrect  @ @ $3p_x$, $3p_y$ $3p_z$ must have energy level
		TOTAL

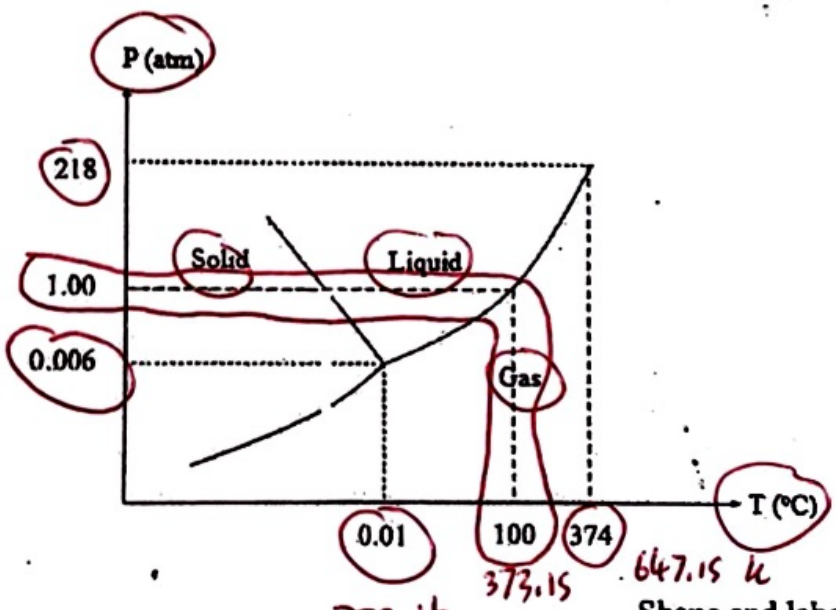
* wrong cancel wright

ma

m_{10}

NO.	PART	SCHEME
3	(a)(i)	Total valence electron for SeO_2 = (1×6) + (2×6) = **18** /m1 octet
	(a)(ii)	VSEPR theory states that a central atom with /M3 ▪ (2 bonding pair and 1 non-bonding pair) therefore ▪ **bent** molecular geometry @ 2 σ-bond or 1 LP or v-shape @ bend or lonepair are accepted  or drawings also accepted
	(a)(iii)	The orbital hybridization is **sp^2**
	(a)(iv)	(✗ da each also accepted) **Each Se=O bond is polar** towards the oxygen atom. **The bonding dipole moments are not cancelled out.** Therefore the molecule is also **polar.** M8 depend on M7 must have both M8 @ μ≠0 @ unsym mo bond dipole cannot cancel each other
	(b)	 Sea of delocalised valence electrons Diagram with label sea of delocalised valence electrons Number of delocalised electrons equal with charge contribute insist Ca^{2+} equation can ▪ Every Ca **atom releases two electrons** and form Ca^{2+} ion. ▪ Metallic bond forms from the **electrostatic attraction between positive ions with the negatively charge sea of valence electrons.** or Ca^{2+} attractive between with sea of elec attractive ✓ elatrostatic forces not accepted ▪ Calcium can conduct electricity due to **free electrons / delocalised electrons.**

NO.	MARK	SCHEME
	(c)	▪ $AlCl_3$ is **covalent** compound. ▪ It has **no free moving electron** to conduct electricity. ▪ Al is a **metal**. ionist ionx insist ▪ Each **Al atom** **releases/contributes** or donate or its **three valence electrons**. ▪ It has **delocalised electrons** to conduct electricity. **Alternative**: ▪ Al is a **metal**. ▪ Has **overlapping** between conduction band and valence band @ @ no forbiden gap **No band gap** between conduction band and valence band. ▪ **Electron can move** from the conduction band to the valence band to conduct electricity.
		TOTAL
		MAXIMUM

NO.	PART	SCHEME
4	(a)	$P_T = P_{H_2} + P_{H_2O}$ @ $P_{H_2} = (0.980 - 0.0313)$ atm $= 0.9487$ atm @ } m1 $PV = nRT$ @ $n_{H_2} = \frac{(0.9487 \text{ atm})(7.80 \text{ L})}{(0.08206 \text{ L atm K}^{-1}\text{mol}^{-1})(298.15 \text{ K})}$ @ } M2 $= 0.302$ mol 0.3 From equation: 1 mol $H_2 \equiv$ 1 mol Zn @ 0.302 mol $H_2 \equiv$ 0.302 mol Zn } comparison m3 Mass of zinc = 0.302 mol × 65.4 g mol^{-1} M4 = 19.78 g M5 unit × insist 19.62
	(b)(i)	 Shape and label axis Label phase, triple and critical points Boiling point line (100 °C, 1 atm) @
	(b)(ii)	Water is **denser than ice** because **melting point** of ice **decreases with pressure.** @ (Density + volume) together water (also accepted) Ice has open structure volume of ice > volu
		TOTAL
		MAXIMUM

$PV = \frac{m}{M_r}RT$ / n4 ecf only if Mr is correct

SOAL	BAHAGIAN	SKEMA
5	(a)	At 400°C = 400 °C + 273.15 K = 673.15 K $K_c = K_p(RT)^{-\Delta n}$ M1 n kecil → $\Delta n = (5 - 4) = 1$ M2 $K_c = (8.3 \times 10^{-3})(0.08206 \times 673.15K)^{-1}$ M3 $K_c = 1.5025 \times 10^{-4}$ M4
	(b)	Mass of HCl will increase. M5 Since this reaction is endothermic, based on Le Chatelier's principle, increasing temperature of the reaction will shift the equilibrium position to the right. M6 forward ✓ system ✗ reaction ✗ equilibrium to the right
	(c)	From (a): $K_c = 1.5025 \times 10^{-4}$ $Q_c = \frac{[HCl]^4[O_2]}{[Cl_2]^2[H_2O]^2}$ M8 $Q_c = \frac{(0.25)^4(0.05)}{(0.70)^2(0.35)^2}$ @ $= 3.254 \times 10^{-3}$ M9 together $Q_c > K_c$ Reaction is NOT at equilibrium. The net reaction proceed to the left. backward reverse ✓
		TOTAL
		MAXIMUM

NO	PART	SCHEME
6	(a)(i)	

	C_6H_5COOH	+ H_2O ⇌	$C_6H_5COO^-$	+ ~~OH⁻~~ H_3O^+
[ ]initial	0.1	–	0	0
[ ]change	$-x$	–	$+x$	$+x$
[ ]equilibrium	$0.1-x$	–	x	x

/m1

only if studs declare HA = C_6H_5COOH

insist

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} = \frac{(x)(x)}{0.1-x} = 3.5 \times 10^{-8}$$

Assume that $x << 0.10$, then $0.10 - x \approx 0.1$

Hence $[H_3O^+] = x = 5.9 \times 10^{-5}$ M

$[A^-] = x = 5.9 \times 10^{-5}$ M

$[HA] = 0.1 - x = (0.10 - 5.9 \times 10^{-5}) = 0.10$ M

if all has no unit – 1 marks

(a)(ii)

$pH = -\log [H_3O^+]$ @

$= -\log (5.9 \times 10^{-5})$ /m1

$= 4.23$

@ $pH = pka + \log \frac{[A^-]}{[HA]}$

+ H_2O ✗ accepted

(b)(i)

$Mg(OH)_2(s) \rightleftharpoons Mg^{2+}(aq) + 2OH^-(aq)$

phase insist arrow

	$Mg(OH)_2$ (s)	⇌	Mg^{2+} (aq)	+ $2OH^-$ (aq)
[ ] initial (M)	–		–	–
[ ] change (M)	–		$+x$	$+2x$
[ ] equilibrium (M)	–		x	$2x$

/m

@ perjantian

$K_{sp} = [Mg^{2+}][OH^-]^2$ @ /m10

$5.61 \times 10^{-12} = (x)(2x)^2$

$4x^3 = 5.61 \times 10^{-12}$

$x = 1.12 \times 10^{-4}$ M /m11

unit insist

NO.	PART	SCHEME
	(b)(ii)	**<u>Addition of $Mg(NO_3)_2$</u>** $Mg(OH)_2\,(s) \rightleftharpoons Mg^{2+}\,(aq) + 2OH^-\,(aq)$ $Mg(NO_3)_2\,(aq) \rightarrow Mg^{2+}\,(aq) + 2NO_3^-\,(aq)$ m12 arrow i phase x **Mg^{2+} is common ion** m13 Based on Le Chatelier's Principle: • **When concentration of Mg^{2+} ion increases, the equilibrium position is shifted to the left** • More $Mg(OH)_2$ precipitate is formed and **solubility of $Mg(OH)_2$ will decrease**
		TOTAL
		MAXIMUM